

*Teaching Notes 5B**

Multiples

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Overview

- A very “short” lecture
- Quickly discuss how sometimes Terminal Values are computed using “Multiples”

Valuation using Multiples

- Valuation by multiples is a “fancy” name for market prices divided by some measure of performance.
- It assesses the value based on that of other (publicly-traded) firms:

$$[\text{Value/Key Characteristic}]_{\text{Ind.Average}} \times \text{Key Characteristic of Firm}$$

- Numerator of multiple is typically the total value of the firm.
- Denominator of multiple is the characteristic that is important for that industry:
 - Clicks or Subscribers For Website
 - Paid Miles Flown for Airlines
 - Number of Patents For a Hi-Tech Firm

Valuation Multiples

- Cash-flow-based **Value** multiples:
 - MV Firm/Earnings, MV Firm /EBITDA, MV Firm /FCF
- Cash-flow-based **Price** multiples:
 - Price/Earnings (P/E), Price/EBITDA, Price/FCF
- Asset-based **Value** multiples:
 - MV Firm/BV Assets, MV Equity/BV Equity
- Industry-specific **Value** multiples:
 - MV Firm/Hospital Beds, MV Firm/Number of Customers

Procedure

Hope: Firms in the same business should have similar multiples.

- **Step 1:** Identify firms in same business as the firm you wish to value.
- **Step 2:** Calculate multiple for comparable firms.
- **Step 3:** Calculate average for the set of comparable firms. In doing this, we are coming up with an estimate of the multiple we wish to use in valuing our firm.
- **Step 4:** Multiply average multiple with the value of the characteristic for the firm you wish to value.
- **Step 5:** Often, different multiples give you different answers; you need to reflect on what is economically reasonable.

Important Remarks

- **Remark I:** When choosing comparable firms, you face a trade-off:
 - Too many firms requires selecting firms that are not truly comparable.
 - Too few firms means that your average will reflect idiosyncrasies of those firms.
- **Remark II:** For firms with no earnings or limited asset base (*i.e.* hi-tech):
 - Price-to-Patents Multiple
 - Price-to-Subscribers Multiple
- **Remark III:** Any differences in characteristics of comparable firms, such as leverage, are difficult to account for
 - For example, we don't know how much value is created from operations and how much is from financing

Takeaways

- Multiples are a quick way to arrive at a valuation
 - Does not mean they are always right (or wrong)
- You should be able to recognize which method is more convenient given the characteristics of the firm.
 - Rapidly growing firms, for example, are difficult to value using WACC or APV.
- As a credibility check on any valuation calculation, it is often useful to try several approaches for estimating a terminal value and hope for some convergence in the results.
 - Trading and transaction multiples provide discipline by ensuring that your valuation is in line with other valuations

Appendix: Additional Notes

Alternative Methods For Estimating Terminal Values

Actual And Pro-Forma Financial Data

		Actual	Pro Forma				
		2017	2018	2019	2020	2021	2022
		Assumed Values					
1	Initial Sales	50.00					
2	End Year PP&E	25.00					
3	End year NWC	20.00					
4	End Year Other Assets	0.00					
5	Annual Growth rate		0.05	0.05	0.05	0.05	0.05
6	COGS/Sales	0.62	0.62	0.62	0.62	0.62	0.62
7	SG&A/Sales	0.20	0.20	0.20	0.20	0.20	0.20
8	Tax Rate	0.40	0.40	0.40	0.40	0.40	0.40
9	NWC/Sales	0.40	0.40	0.40	0.40	0.40	0.40
10	PP&E/Sales	0.50	0.50	0.50	0.50	0.50	0.50
11	Dep/PP&E	0.10	0.10	0.10	0.10	0.10	0.10
12	Other Assets/Sales	0.00	0.00	0.00	0.00	0.00	0.00
13	Discount Rate	0.10					
		Output Values					
14	Sales	50.00	52.50	55.13	57.88	60.78	63.81
15	End Year PP&E	25.00	26.25	27.56	28.94	30.39	31.91
16	End Year NWC	20.00	21.00	22.05	23.15	24.31	25.53
17	End Year Other Assets	0.00	0.00	0.00	0.00	0.00	0.00
18	End Year Total Capital	45.00	47.25	49.61	52.09	54.70	57.43

Computing Free Cash Flows

$$FCF_t = EBIT_t(1 - t) + Dep_t - CAPX_t - \Delta NWC_t$$

	Free Cash Flow	2017	2018	2019	2020	2021	2022
19	Net Sales	50.00	52.50	55.13	57.88	60.78	63.81
20	COGS	31.00	32.55	34.18	35.89	37.68	39.56
21	SG&A	10.00	10.50	11.03	11.58	12.16	12.76
22	Depreciation	2.50	2.63	2.76	2.89	3.04	3.19
23	EBIT	6.50	6.83	7.17	7.52	7.90	8.30
24	EBIT(1-t)	3.90	4.10	4.30	4.51	4.74	4.98
25	Plus: Depreciation		2.63	2.76	2.89	3.04	3.19
26	Minus: Change In NWC		1.00	1.05	1.10	1.16	1.22
27	Minus: CAPEX		3.88	4.07	4.27	4.49	4.71
28	Free Cash Flow		1.85	1.94	2.03	2.14	2.24

- Additional Information:
 - Book Value at the end of year 2017 is \$45M.
 - Debt = \$15M, Equity = \$30M
 - 1M common shares outstanding and stock price = \$26 p/share
 - Book Value p/share = \$30
 - Market Value Equity/Book Value Equity = \$26/\$30 = 0.87

Five Approaches To Computing Terminal Values

- With no terminal value calculation and a 10% discount rate, the PV of the cash flows is **\$7.66M**.
- There are **five approaches** to computing Terminal Values:
 - **Approach I:** TV as a Growing Perpetuity Cash Flow
 - **Approach II:** TV as a Stable Perpetuity Cash Flow
 - **Approach III:** TV using Salvage Value
 - **Approach IV:** TV as a Multiple of Book Value
 - **Approach V:** TV as a Multiple of Earnings

Approach I: Growing Perpetuity Cash Flow

- Suppose cash flows continue to grow at the rate of 5% per year in the future.
- If you do a perpetuity calculation, you will probably use the last free cash flow and divide it by $[WACC - g]$.
 - Terminal FCF in Last Year T. The main issue is to determine the FCF in a steady-state situation.
 - Be careful with fast-growing companies or cyclical industries.
- Get FCF_T and then apply formula:

$$TV_T = \frac{FCF_T(1 + g)}{WACC - g}$$

- Where do we get g ?
 - More room for skewing your answer with this choice.

Approach I: Growing Perpetuity Cash Flow (ctd.)

- To get PV(TV), need to discount back:

$$PV(TV) = TV_T / (1 + WACC)^T$$

- Make sure FCF_T used to get TV reflects g :
 - Adjust capital expenditures and depreciation.
 - Adjust change in net working capital.
 - Careful with asset sales and extraordinary income.

	2017	2018	2019	2020	2021	2022	2018-2022 Totals
Free Cash Flow		1.85	1.94	2.03	2.14	2.24	
PV Factor @ 10%		0.909	0.826	0.751	0.683	0.621	
PV(No Terminal Value)		1.68	1.60	1.53	1.46	1.39	7.66
Approach I							
TV as Growing Perpetuity						47.09	
PV(TV)							29.24
Total Enterprise Value							36.90
Minus: Book Value Debt							15.00

Approach II: Stable Perpetuity Cash Flow

- If we assume no growth in sales after year 5, then FCF beyond 2022 will exactly equal $EBIT(1 - t)$.
 - No new funds will be required to support additional sales, and CAPX will exactly offset depreciation.
 - Capital will not need to grow anymore.
 - Get FCF_T , apply present value factor, and compute $PV(TV)$:

$$TV_T = \frac{FCF_T}{WACC} \quad PV(TV_T) = \frac{TV_T}{(1 + WACC)^T}$$

Approach II		2018	2019	2020	2021	2022	2018-2022 Totals
TV as Stable Perpetuity	FCF = EBIT(1-t)					49.77	
PV(TV)							30.91
Total Enterprise Value							38.56
Minus: Book Value Debt							15.00
Market Value Equity							23.56

Approach III: Terminal Value with Salvage Value

- Assume a **salvage value** for the assets of the business at the end of the last year of the planning horizon.
- Liquidation Analysis:
 - The 2022 year-end NWC might be liquidated at cost for a total recovery of **\$25.5M**.
 - PP&E might be liquidated at 50% of net book value, therefore $0.5 \times \$31.91 = \mathbf{\$15.96M}$

Approach III							
Liquidation Analysis		Book Value		% Recovery		\$ Recovery	
		Dec-22					
NWC		25.53		100%		25.53	
PP&E		31.91		50%		<u>15.95</u>	
Subtotal		57.43				41.48	
Tax Benefit Write-Off PP&E						6.38	
						<u>47.86</u>	
PV(TV)							29.72
Total Enterprise Value							37.38
Minus: Book Value Debt							15.00
Market Value Equity							22.38

Approach IV: Multiple Of Book Value Capital

- Multiply the forecasted terminal-year book value of invested capital by an appropriate **Market Value/Book Value** ratio.
- The **current** Market Value/Book Value for the business is generally utilized.
- In our example, using the information from earlier:
 - Market Value Total Capital = $\$26 \times 1\text{M} + \$15\text{M} = \$41\text{M}$
 - Book Value Total Capital = $\$30\text{M} + \$15\text{M} = \$45\text{M}$
 - Market/Book Ratio Total Capital = $\$41/\$45 = \mathbf{0.91}$

Approach IV		Market		Book		Market/Book	
Debt		15.00		15.00		1	
Equity		26.00		30.00		0.87	
Total Capital		41.00		45.00		0.91	
TV as Market/Book Ratio	= Market/Book * Total Capital					52.3	
PV(TV)							32.49
Total Enterprise Value							40.15
Minus: Book Value Debt							15.00
Market Value Equity							25.15

Approach V: Multiple Of Earnings

- Multiply the forecasted terminal-year profits by an appropriate **Price/Earnings** ratio.
- The **current** Price/Earnings Ratio for the business is generally utilized.
- The market value of the firm's total capital is \$41M and total profits in 2017 were \$3.9M.
 - Price/Earnings Ratio = $\$41/\$3.9 = 10.5x$

Approach V				P/E			
TV as a Multiple Of Earnings	= P/E * EBIT(1-t)			10.5		52.33	
PV(TV)							32.49
Total Enterprise Value							40.15
Minus: Book Value Debt							15.00
Market Value Equity							25.15

Summary Of Results

- These are the results for the 5 different approaches to computing terminal values:

	TV	Present Value			
	2022	TV	Debt	Equity	Enterprise
Growing Perpetuity	47.09	29.24	15.00	21.90	36.90
Stable Perpetuity	49.77	30.91	15.00	23.56	38.56
Multiples of Book Value	52.33	32.49	15.00	25.15	40.15
Multiples of After-Tax Earnings	52.33	32.49	15.00	25.15	40.15
Liquidation	47.86	29.72	15.00	22.38	37.38

- The 5 calculated values for the outstanding common shares of the firm are all below the \$26M market value at which the shares are currently trading.
 - This gap could imply that investors do not expect the firm to continue “business as usual”.

“Other Multiples”

Transaction Multiples

- Another way to value comparable firms is to use the value that was paid for comparable firms in a **merger** or **acquisition**.
- The steps involved are identical to the Trading Multiples Procedures, except that this method focuses solely on firms that were **acquired**.
 - The Enterprise Value of the firm is the amount paid for the equity, plus debt, by the acquirer!
- Transaction multiples often **exceed** trading multiples. Why?
 - A higher transaction multiple implies a higher $1/(r - g)$. For the same r , this implies a higher g .
 - Often, acquirers argue that the g of cash flows is higher due to synergies between the acquirer and the target.

Important Remark

- **Remark:** Be careful when using P/E ratios to compute Equity Value, especially if the comparables have different leverage!
- P/E ratios have **two** major flaws.
 - They are systematically affected by capital structure! P/E ratios tend to rise with leverage. Companies can artificially increase P/E ratio by swapping debt-for-equity.
 - They are based on earnings, which include many non-operating items, such as restructuring charges and write-offs. Since these are often one-time events, multiples based on P/Es can be misleading.

Motivation for Cash Flow Multiples?

- Firms in the same business should have similar multiples.
- If the firm's actual FCF is a perpetuity:

$$\text{MV Firm} = \text{FCF}/(\text{WACC} - g) \rightarrow \text{MV Firm}/\text{FCF} = 1/(\text{WACC} - g)$$

- Comparables will have a similar MV Firm/FCF provided they:
 - Have the **same** WACC (requires similar $D/(D+E)$).
 - Grow at a **similar** rate.
- **Remark:** The cash flow multiple is useful when the firm is in a period of stable growth and margins.
- **Note:** Since these are rough approximations (at best), one may want to check if other multiples capture systematic deviation. In particular, see if some multiples are quite constant across firms.

Motivation for EBITDA and Sales Multiples?

- The EBITDA or Sales Multiples do not have a direct relation to the NPV formula.
 - $FCF = EBITDA(1 - t) + t \times Dep - CAPX - \Delta NWC$
 - During the 1990s, EBITDA multiples became a popular way of valuing firms. Firms had negative FCFs!
- If the RHS terms of the FCF equation ($t \times Dep - CAPX - \Delta NWC$) are small or proportional to EBITDA, then multiplying FCF by a constant is the **same** as multiplying EBITDA by a constant.
 - We can ignore fluctuations in Depreciation, CAPX, and NWC, which can have strong effects on value if we use a FCF multiple!
- Same logic has been applied to explain the applicability of sales multiples or customer multiples.

Motivation for EBITDA and Sales Multiples? (cont.)

- If the ultimate cash flows of the business are proportional to the stream of sales (or customers), then it may be appropriate to value the firm as a constant times this year's sales.
- **Caveat:** The assumed relationship between sales (or customers) and cash flows **may** or **may not** hold!
- Following the crash of internet stocks in 2002, these alternative approaches to valuation fell out as investors realized that sales and EBITDA were not always indicative of future cash flows.
- **Bottom Line:** Warren Buffet, a long-time critic of EBITDA multiples, suggested:
 - “Does management think the tooth fairy pays for capital expenditures?”

Pros and Cons of Multiples

- **Pros:**

- Incorporates a lot of information from other valuations in a simple way.
- Embodies market consensus about discount rate and growth rate.
 - Free-ride on market's information.
- Can provide discipline in valuation process by ensuring that your valuation is in line with other valuations.
- Sometimes, what you care about is what the market will pay, not the fundamental value.

Pros and Cons of Multiples (cont.)

- **Cons:**

- Difficult to find **true** comparables. Implicitly assumes all comparables are alike in growth rates, cost of capital, and business composition.
- Hard to incorporate firm-specific information. Particularly problematic if operating changes are going to be implemented.
- Relies on accounting measures being comparable too.
- Differences in accounting practices can affect earnings and equity-based multiples → Better to use cash flow multiples.
- Book values can vary across firms depending on age of PPE.
- Multiples are backward-looking, valuation should be **forward-looking**!
- If everyone uses comparables, who actually does fundamental analysis? That is, if market is overpaying, you will too!